Project Report

On

Cab Booking Application

Developed as part of the LNCT Group Placement Initiative for the SkillWallet SmartBridge Program.

Developed By:

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Under the Guidance of

SmartBridge Experiential Learning Program

LNCT Group of Colleges, Bhopal

Approved by AICTE New Delhi & Govt. of M.P.

Affiliated to Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P.)

Year - 2026

CERTIFICATE

This is to certify that the project titled "CAB BOOKING APPLICATION – Online Cab Booking System" has been successfully completed by

Prince Raj Singh

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Rimjhim Singh

as a part of the Skill Wallet – SmartBridge Program under the Placement Initiative of LNCT Group of Colleges, Bhopal.

This project work is submitted as a part of practical learning and project development in the field of Computer Science and Engineering. The project has been carried out under proper guidance and supervision and demonstrates the practical implementation of modern web technologies including frontend, backend, and database integration.

Project Guide / Mentor

(Signature)

DECLARATION

We hereby declare that the project titled **"Cab Booking Application"** is an original work carried out by me as part of the **Skill Wallet – SmartBridge Program** under the placement initiative of **LNCT Group of Colleges, Bhopal**.

This project has been developed with the objective of applying practical knowledge of **Web Development, Database Management, and System Design**. The work presented in this documentation is entirely based on my own efforts and has not been submitted previously for any other certification or academic award.

All the resources, references, and tools used in this project have been properly acknowledged.

We take full responsibility for authenticity and originality of this work.

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Date:

Program: Skill Wallet – SmartBridge Program

Project Title – Cab Booking Application

Signature -

ACKNOWLEDGMENT

WeI would like to express my sincere gratitude to the **LNCT Group of Colleges** and the **SmartBridge Program Team** for providing me with the opportunity to work on the project titled **"cab Booking Application"**.

This project has been a great learning experience and has helped me enhance my technical knowledge and practical skills.

I would also like to thank my mentors and coordinators for their valuable guidance, continuous support, and encouragement throughout the development of this project.

Finally, I thank everyone who directly or indirectly supported me in completing this project successfully.

Project Team

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1. Project Title

Cab Booking Application

2. Abstract

Cab Booking Application is a web-based application developed to simplify transportation services by allowing users to book cabs online. The system provides a convenient platform where users can register, log in, and book rides between different locations.

The application is developed using modern web technologies such as **React.js for frontend, Node.js and Express.js for backend, and MongoDB for database management**.

The system helps users quickly find available cabs, manage bookings, and maintain ride records digitally. This project demonstrates how technology can improve urban transportation by making cab booking more efficient and accessible.

3. Introduction

3.1 Background of the Project

Transportation plays an important role in daily life. Many people rely on cab services to travel from one place to another.

Traditional cab booking methods often involve calling drivers or searching manually, which can be time-consuming and inconvenient.

With the advancement of digital technology, online cab booking systems provide a faster and more efficient way to manage transportation services.

3.2Problem Statement

- Difficulty in finding available cabs
- Time-consuming manual booking
- Lack of ride tracking
- Poor management of booking records
- Inconvenient communication with driver

3.3 Proposed Solution

The proposed system **Cab Booking Appllication** provides an online platform where users can:

- Register and create accounts
- Log in securely
- Enter pickup and drop locations
- Book cabs instantly
- View booking details

This system simplifies the booking process and improves the overall travel experience.

4. Objectives of the Project

The main objectives of the Cab Booking Application project are:

- To develop an online cab booking system
- To provide a simple and user-friendly interface
- To make cab booking faster and easier
- To maintain digital records of rides
- To improve transportation service management

5. Scope of the Project

The scope of the project includes:

- User registration system
- Secure login system
- Cab booking interface
- Ride history management
- Database storage of booking details

Future enhancements may include:

- Online payment integration
- Live GPS tracking
- Driver rating system
- Mobile application
- Real-time cab availability

6. Literature Review

Many modern transportation systems such as online taxi services provide digital platforms for booking rides.

These systems use web and mobile technologies to manage ride bookings efficiently.

Studies show that online booking systems improve convenience, reduce waiting time, and provide better management of transportation services.

7. System Requirements

7.1 Hardware Requirements

- Computer or Laptop
- Minimum 4GB RAM
- Internet connection

7.2 Software Requirements

- Operating System: Windows / Linux
- Frontend: React.js
- Backend: Node.js
- Framework: Express.js
- Database: MongoDB
- Code Editor: VS Code
- Browser: Chrome / Edge

8. System Architecture

The Cab Booking Application system follows a **three-tier architecture**.

1. Presentation Layer (Frontend)

This layer provides the user interface where users interact with the system.

2. Application Layer (Backend)

This layer processes user requests and handles application logic.

3. Database Layer

This layer stores user data and ride booking information.

Flow:

User → Frontend → Backend → Database → Backend → Frontend → User

9. Modules of the System

9.1 User Registration Module

Allows new users to create accounts by entering details such as name, email, and password.

9.2 Login Module

Allows registered users to log in securely using their credentials.

9.3 Cab Booking Module

Users can enter pickup and drop locations and book a cab.

9.4 Admin Module

Admin can manage users, bookings, and cab records.

10. Database Design

The system uses MongoDB, a NoSQL database.

Main collections include:

Users Collection

- UserID
- Name
- Email
- Password

Bookings Collection

- BookingID
- UserID
- PickupLocation
- DropLocation
- Date
- RideStatus

11. ER Diagram (Description)

The Entity Relationship Diagram contains two main entities:

User Entity

- UserID
- Name
- Email
- Password

Booking Entity

- BookingID
- PickupLocation
- DropLocation
- Date
- RideStatus

Relationship: One user can make multiple Cab bookings.

12. Implementation

The project implementation consists of three main parts:

Frontend Implementation

Developed using React.js to create interactive user interfaces.

Backend Implementation

Implemented using Node.js and Express.js to handle server-side logic.

Database Implementation

MongoDB stores user and booking data.

13. Testing

Testing ensures that the system functions correctly.

Types of testing performed:

- Unit Testing
- Integration Testing
- System Testing

Testing helps identify bugs and ensures system reliability.

14. Advantages of the System

- Easy cab booking
- Saves time
- Digital record management
- User friendly interface
- Efficient ride management

15. Limitations

- Requires internet connection
- Limited to registered users
- Depends on server availability

16. Future Scope

Future improvements may include:

- Online payment system
- GPS based tracking
- Driver rating and feedback system
- Mobile application support
- Real-time driver availability

17. Conclusion

Cab Booking Application provides a modern solution for managing transportation services. The system allows users to book cabs easily and manage ride details digitally.

By using modern web technologies, the system simplifies the process of cab booking and improves the overall user experience.